

data is fed into the APS computer of the HK-APS.

If the computer finds that the trajectory of the projectile is towards the tank, it calculates the right time and angle to destroy the RPG. Once the RPG has reached the right area, the control computer activates the rocket launcher to fire a short-range interceptor rocket. This rocket, after reaching the proximity of the RPG, explodes and releases many fragments that in turn destroy the RPG. The same methodology applies for anti-tank missiles also.

The reaction time of these hard-kill active protection systems is generally in milliseconds after an RPG is detected. The RPG is intercepted within a blink of an eye. After destroying an RPG, the system gets ready to destroy the next projectile within half a second. In a separate interactive display, the tank commander sets the modes of operation and monitors status of the system. Even if a projectile passes without harming the tank, through this display the HK-APS indicates to the tank commander the direction from where it was fired. The tank commander may, in turn, attack that source of attack.

The design of these APS is such that they do not respond to decoys. The modern Russian HK-APS systems are rumoured to be even capable of working in a group with the APS of the adjacent tanks. But further details are kept as highly classified information. In general, how the APS distinguishes between different types of missiles and accordingly fixes its *modus operandi* are not revealed in the open domains.

German AMAP-ADS, Swedish LEDS-150 and American Quick Kill are some of the HK-APS systems in pre-induction phases, while Israeli Iron Fist and Trophy and Russian Arena are fully operational. The only disadvantage of HK-APS is that it is dangerous for the infantry to be within 20-30 metres of the tank during joint infantry-armour operations. A tank can either have SK-APS or HK-APS, not both.

But APS cannot offer protection against the high-speed anti-tank shells fired by other tanks. Because of the



Fig. 9: An Iraqi soldier with Russian rocket-propelled grenade-7 (RPG-7) (Photograph credit: US Marines, through Wikipedia)



Fig. 11: Interceptor launcher of Israeli 'Iron Fist' APS (Photograph credit: Ereshkigal1 through Wikipedia)

speed (1.5 km/s) at which these shells travel, it is impossible for any system to intercept them. The best way to avoid these shells is through better tactics so as to avoid getting shot at. But in spite of the tactics if a tank is shot, the armour is the last line of defence. Notwithstanding the armour's strength, electronics has penetrated these armours. So, now smart armours are being developed to replace the traditional armours.

Smart armour

The armour explained below, though still in research phase in the UK, is likely to enter the battlefield in the next decade. This explosive reactive armour (ERA) uses layers of highly explosive material sandwiched between armour plates. When a projectile hits the ERA, the hit detonates the explosive layers. The resulting explosion pushes the outer steel plates on to the warhead and disrupts its flow. All this happens in milliseconds. This counter explosion

reduces the force of impact greatly and saves the tank crew. This armour is presently used in the form of ERA tiles fitted over the existing tank's armour. The ERA tiles have to be replaced after a hit.

The 'smart' concept has been extended to the ERA. In future, the ERA will be formed as small ERA blocks instead of ERA tiles and their detonation will be controlled by a computer. Pressure sensors will be embedded into the ERA blocks. On a shell's hitting the armour, the computer through the pressure sensors would sense the location, velocity and diameter of the shell from the impact. Accordingly, the computer would detonate only the relevant explosive elements.

Through this 'Smart ERA' concept, minimum force will be used to destroy the shell. Further, the need to change an entire ERA tile would be eliminated. The smaller damaged explosive blocks would be easily replaced at a later time. Since there is only one-in-a-million chance of two shells hitting at the same point, the tank will be safe even without an immediate ERA replacement.

To sum up, electronic tank protection systems have become the tank's first line of defence, pushing the armour to be the last resort. Army strategists, due to these APS, foresee the reduction of armour weight. Such reduction will make the tanks light, fuel-efficient and agile.

Electronic systems to defend tanks are fine but what about the electronic systems for offensive capabilities? For that the electronic fire control system comes into the play, which will be described in the next part.

To be continued next month

The author has contributed several articles in the past as well